

StudyBuddy — Nuri Parent Hub Specification

AI-Powered Parent Interface (Beyond Twinkl)

Version: 1.0

Extends: studybuddy-spec.md (Section 15: Parent Highlights → replaced by Parent Hub)

Date: March 31, 2026

1. Product Vision

The Parent Hub is Nuri's parent-facing interface — a companion app/section that gives parents superpowers to support their child's learning. Unlike Twinkl (which sells static, one-size-fits-all worksheets), Nuri's Parent Hub generates **personalized, AI-powered resources** based on each child's actual performance, curriculum, year level, and weak spots.

The headline: "Twinkl gives every Year 3 student the same fractions worksheet. Nuri gives YOUR child a worksheet focused on the exact fractions they got wrong last week."

2. Core Philosophy

- Parents want to help but often don't know HOW or WHERE their child is struggling
- Parents shouldn't need to be teachers — Nuri does the heavy lifting
- Every resource is personalized to THIS child, not generic
- Arabic + English bilingual resources generated on demand — no other platform does this
- Parents feel informed, not surveilled; children feel supported, not watched
- Printable resources complement (not replace) screen-based learning

3. Parent Hub Features

3.1 Smart Dashboard (Daily Overview)

The landing page parents see when they open the Parent Hub.

What's Displayed:

[illegible]

- "How did I do?" self-assessment section at the bottom

Generation Logic:

System prompt includes:

- Child's year level
- Child's placement assessment results
- Last 20 quiz results for this topic
- Specific error patterns detected
- Mistake Journal entries for this topic

Example: If error pattern = "conceptual: adding denominators"

- Generate questions that specifically require finding common denominators
- Include 2 worked examples showing WHY you can't add denominators
- Progress from simple (same denominator) to complex (different denominators)
- Include a "spot the mistake" question where the mistake matches their error pattern

PDF Output Styling:

- A4 and US Letter size options
- Nuri branding: owl mascot, orange-purple gradient header
- Clear, child-friendly font (Nunito)
- For Arabic worksheets: RTL layout, Noto Naskh Arabic font, proper tashkeel
- Bilingual worksheets available: Arabic on right side, English scaffolding on left

Technical Implementation:

- Claude API generates the content (questions, examples, answers)
- ReportLab (Python) or react-pdf renders the PDF
- Cached templates for common layouts (maths grid, reading comprehension, Arabic practice)
- Generated PDFs stored temporarily (7 days) for re-download

3.3 Bilingual Arabic + English Resources ■

What makes this unique: No competitor generates bilingual Arabic/English educational resources dynamically.

Resource Types:

Resource	Description	Example
Arabic Vocabulary Cards	Flashcard-style printables with Arabic word + transliteration + English + image	■■■■■■■ — kitaab — book — [book image]
Arabic Reading Passages	Short passages at the child's Arabic level with comprehension questions in English	Year 3 Arabic passage with "What happened?" questions in English
Bilingual Worksheets	Side-by-side Arabic/English for any subject	Maths word problems: Arabic on right, English on left
Tashkeel Practice Sheets	Arabic words without vowel marks — child adds them	■■■■■■■ practice matching the child's year level
Grammar Worksheets	■■■■■■■■ ■■■■■■■■, ■■■■■■■■ etc. with English explanations	Arabic grammar at their ACTUAL level (not school year)
Religion Study Sheets	Key terms, prayers, Bible passages in Arabic + English	■■■■■■■■■ — al-Qiyama — Resurrection + context

Smart Arabic Level Handling:

- Uses Placement Assessment data: child might be Year 6 in school but Year 3 in Arabic
- All Arabic resources generated at their ACTUAL Arabic level, not school year
- English scaffolding adjusts: more English support for weaker Arabic readers
- Tashkeel (vowel marks) included for Years 1-3, gradually removed for Years 4+
- Transliteration included for Years 1-4, optional for Years 5-6

3.4 Smart Weekly Practice Plan ■

Every Sunday, Nuri auto-generates a personalized weekly practice plan based on the child's data.

What Gets Analyzed:

- Quiz performance over the past 2 weeks
- Topics due for spaced repetition review
- Mistake Journal unresolved items
- Confidence Meter blind spots (confident but wrong)
- Upcoming test prep (if any Pre-Test Predictor is active)
- Subject balance (did they skip any subject last week?)

Generated Plan Example:

■ Carla's Week – March 31 to April 6

- Last Week: Studied 5/6 subjects (skipped History)
 - 72% quiz accuracy (target: 80%)
 - 9 unresolved mistakes

This Week's Focus:

[illegible]

Monday: Maths – Equivalent Fractions (review mistakes)
 ■ [Download Practice Sheet: 15 questions]

Tuesday: Arabic – Sun & Moon Letters (spaced repetition due)
 ■ [Download Vocabulary Cards: 20 words]

Wednesday: Science – Water Cycle (new topic from class)
 ■ Suggest: Learn Mode session (15 min)

Thursday: History – Ancient Egypt (skipped last week!)
 ■ Suggest: Quick quiz (10 questions)

Friday: Maths – Fractions review + English spelling test
 ■ [Download Bilingual Worksheet]

Weekend: Free choice! Nuri recommends Story Mode Chapter 3 ■

Daily: 5 min Arabic reading practice (reading level building)

Parent Controls:

- Can adjust the plan (swap days, skip subjects)
- Set preferred study days and times
- Mark items as "done at school" (removes from plan)
- Add custom practice requests ("extra times tables this week")

3.5 Homework Photo → Parent Explainer ■

Already in the spec as Snap & Learn, but with a parent-specific mode.

Child Mode (existing): Child photographs homework → Nuri guides them through it Socratically (never gives answers)

Parent Mode (new): Parent photographs the same homework → gets a clear, adult-level explanation of:

- **What the question is asking** (decoded from the photo)
- **The concept behind it** (explained for an adult, not a child)
- **How to explain it to your child** (age-appropriate analogy suggestions)
- **Common mistakes to watch for** (from Error Pattern Detection data)
- **The correct answer with working** (so the parent can verify)

Example:

Parent scans a Year 4 fractions worksheet.

■ Parent Explainer

Question: "Write $\frac{3}{4}$ as an equivalent fraction with denominator 12"

■ The Concept:

Equivalent fractions are different fractions that represent the same amount. To convert $\frac{3}{4}$ to a denominator of 12, you multiply both the numerator and denominator by the same number ($12 \div 4 = 3$, so multiply both by 3).

Answer: $\frac{3}{4} = \frac{9}{12}$ (because $3 \times 3 = 9$ and $4 \times 3 = 12$)

■■ How to explain to Carla:

"Imagine cutting a pizza into 4 slices and eating 3. Now imagine cutting the SAME pizza into 12 smaller slices – you'd eat 9 of the smaller slices, but it's still the same amount of pizza!"

■■ Watch out:

Carla tends to multiply only the numerator and forget the denominator. Remind her: "Whatever you do to the bottom, you must do to the top!"

Language Support:

- Works for Arabic homework too
- Parent explanation always in English (parents' stronger language)
- Shows the Arabic text from the homework with transliteration
- Explains Arabic grammar concepts in English with clear examples

3.6 Personalized Printable Activity Packs ■

Auto-generated weekly activity packs that parents can print for offline practice, car journeys, or screen-free time.

Pack Contents (auto-curated per child):

Component	Pages	Description
Cover page	1	"{Child's name}'s Activity Pack — Week of April 1" with Nuri illustration
Maths practice	2-3	Problems focused on current topics + review of weak areas
Reading passage	1-2	English reading comprehension at their level
Arabic practice	1-2	Vocabulary, reading, or grammar at their ACTUAL Arabic level
Word search / crossword	1	Vocabulary from current curriculum topics
Fun facts page	1	Science or History facts connected to what they're studying
Coloring / drawing page	1	For younger years (Y1-Y3): topic-related coloring
Answer key	1-2	Separate pages parents can keep
Total	8-12 pages	Personalized PDF, ready to print

Generation Frequency:

- Auto-generated every Sunday
- Parents notified: "Carla's new activity pack is ready! ■"
- Can also generate on-demand packs for specific subjects/topics
- Holiday packs: longer packs for school breaks (20+ pages, revision-focused)

3.7 Progress Reports ■

Two types:

Daily Nuri's Note (Push Notification)

Short, specific, actionable. Sent at a time the parent configures (default: 7pm).

- Today Carla practiced for 14 minutes:
 - Maths quiz: 7/10 (struggled with division word problems)
 - Arabic: reviewed 12 vocabulary cards
- Tip: "Division word problems confuse her because she doesn't know which number to divide by which. Try asking: 'Who is sharing? How many people are sharing?' to help her set up the problem."
- Streak: 13 days

Weekly Summary Report (Email or In-App)

- Nuri's Weekly Report — Week of March 24
- STUDY TIME: 1hr 42min (target: 1hr 45min ■)
 SUBJECTS: 5 of 6 covered (skipped: History)
 QUESTIONS: 87 answered (71% correct — up from 68% last week ■)
 XP EARNED: 340 | Level: 7 → 8 ■
 STREAK: ■ 12 days

SUBJECT BREAKDOWN:

- ★ Science: 89% accuracy (strongest subject!)
- ★ Maths: 74% accuracy (improving – was 65% last month)
- ▲ English: 72% accuracy (steady)
- ▲ Religion: 70% accuracy
- Arabic: 52% accuracy (needs attention)
- ✗ History: not practiced this week

MISTAKES RESOLVED: 9 of 14 (5 still in review queue)

BLIND SPOTS DETECTED:

- Confident but wrong on: place value in 4-digit numbers
- Lucky guesses on: apostrophes for possession

RECOMMENDATION:

"Focus 10 minutes on Arabic daily this week. Carla's listening comprehension is strong (Year 4 level) but her reading is at Year 2. I've added Arabic reading cards to this week's practice plan."

[Download This Week's Activity Pack] [View Detailed Analytics]

Term Report (End of Each School Term)

Comprehensive PDF report covering:

- Overall progress across all subjects
- Placement assessment comparison (start of term vs now)
- Topics mastered and topics still in progress
- Learning style profile update
- Nuri's recommendations for next term
- Graphs showing accuracy trends over time

3.8 Parent Learning Corner ■

Short, digestible content that helps parents understand what their child is learning and how to support them.

Content Types:

Type	Example	Format
"What they're learning this week"	"This week in Year 4 Maths, your child is studying equivalent fractions. Here's what that means and why it matters..."	2-minute read
"How to help at home"	"5 ways to practice fractions at home without any worksheets — using pizza, chocolate bars, and LEGO"	Tips card
"Understanding the curriculum"	"Cambridge Maths vs Egyptian Maths: what's the same and what's different in Year 3"	Article
"Arabic for non-Arabic parents"	"Your child is learning ■■■■■■ ■■■■■■ — here's what that means in English and how to help"	Explainer

Type	Example	Format
"When to worry, when to relax"	"Your child's Arabic is 2 years below school level — here's why that's normal for international school kids and what to do"	Reassurance guide

Generation: AI-generated based on the child's current curriculum and placement level. Proactively triggered when the child starts a new topic or when a significant gap is detected.

4. Parent Hub vs Twinkl — Feature Comparison

Feature	Twinkl	Nuri Parent Hub
Worksheet library	1M+ static resources	∞ AI-generated, personalized per child
Personalization	None — same worksheet for every child	Based on child's performance, mistakes, and level
Arabic resources	Limited, static	Full bilingual generation with tashkeel, transliteration
Egyptian curriculum	No	Yes — Religion, Arabic Language
British curriculum	Yes (strong)	Yes
Progress tracking	No	Full dashboard with daily/weekly/term reports
Smart recommendations	No	AI suggests what to practice based on data
Homework help for parents	No	Photo scan → parent-level explanation
Price	£5-15/month for static library	Included in Nuri subscription
Requires teacher expertise	Yes — parent must pick the right resource	No — Nuri picks for you

Key advantage: Twinkl requires parents to know what their child needs and find the right resource from 1M+ options. Nuri knows what the child needs and generates the exact right resource automatically.

5. Technical Architecture

5.1 API Endpoints (New)

```
# Parent Hub Dashboard
GET    /api/parent/dashboard/:profileId    — Daily overview, tips, stats
GET    /api/parent/weekly-report/:profileId — Weekly summary data

# Worksheet Generator
POST   /api/parent/worksheet/generate      — Generate personalized worksheet
GET    /api/parent/worksheet/:id/pdf       — Download generated PDF
GET    /api/parent/worksheets/:profileId — List generated worksheets

# Activity Packs
POST   /api/parent/activity-pack/generate  — Generate weekly activity pack
GET    /api/parent/activity-pack/:id/pdf — Download activity pack PDF
GET    /api/parent/activity-packs/:profileId — List available packs
```

```

# Practice Plan
GET    /api/parent/practice-plan/:profileId    - Get weekly practice plan
PUT    /api/parent/practice-plan/:id          - Modify practice plan

# Homework Explainer
POST   /api/parent/homework/explain           - Upload photo + get parent explanation

# Progress Reports
GET    /api/parent/reports/daily/:profileId    - Daily Nuri's Note
GET    /api/parent/reports/weekly/:profileId   - Weekly summary
GET    /api/parent/reports/term/:profileId     - Term report PDF

# Learning Corner
GET    /api/parent/learning-corner/:profileId  - Curated articles for parent

```

5.2 Database Additions

```

-- Generated worksheets
CREATE TABLE worksheets (
  id SERIAL PRIMARY KEY,
  profile_id INT REFERENCES profiles(id),
  subject TEXT NOT NULL,
  topic TEXT NOT NULL,
  difficulty TEXT DEFAULT 'at_level',
  num_questions INT,
  content JSONB, -- questions, answers, worked examples
  pdf_url TEXT,
  generated_at TIMESTAMP DEFAULT NOW(),
  expires_at TIMESTAMP DEFAULT NOW() + INTERVAL '7 days'
);

-- Activity packs
CREATE TABLE activity_packs (
  id SERIAL PRIMARY KEY,
  profile_id INT REFERENCES profiles(id),
  week_start DATE,
  pack_type TEXT DEFAULT 'weekly', -- weekly, holiday, on_demand
  content JSONB,
  pdf_url TEXT,
  downloaded BOOLEAN DEFAULT FALSE,
  generated_at TIMESTAMP DEFAULT NOW()
);

-- Practice plans
CREATE TABLE practice_plans (
  id SERIAL PRIMARY KEY,
  profile_id INT REFERENCES profiles(id),
  week_start DATE,
  plan JSONB, -- daily breakdown with tasks and resources
  parent_modifications JSONB, -- parent overrides
  items_completed INT DEFAULT 0,
  items_total INT,
  generated_at TIMESTAMP DEFAULT NOW()
);

-- Parent homework scans
CREATE TABLE parent_homework_scans (
  id SERIAL PRIMARY KEY,
  profile_id INT REFERENCES profiles(id),
  image_url TEXT,
  subject TEXT,
  topic TEXT,
  parent_explanation JSONB, -- concept, how_to_explain, common_mistakes, answer

```

```

    scanned_at TIMESTAMP DEFAULT NOW()
);

-- Parent notifications
CREATE TABLE parent_notifications (
    id SERIAL PRIMARY KEY,
    profile_id INT REFERENCES profiles(id),
    parent_id INT REFERENCES parents(id),
    type TEXT, -- daily_note, weekly_report, pack_ready, milestone
    content JSONB,
    read BOOLEAN DEFAULT FALSE,
    sent_at TIMESTAMP DEFAULT NOW()
);

-- Parent accounts
CREATE TABLE parents (
    id SERIAL PRIMARY KEY,
    name TEXT,
    email TEXT,
    phone TEXT,
    pin_hash TEXT,
    notification_time TIME DEFAULT '19:00',
    notification_method TEXT DEFAULT 'push', -- push, email, whatsapp
    children INT[], -- array of profile_ids
    created_at TIMESTAMP DEFAULT NOW()
);

```

5.3 PDF Generation Pipeline

1. Parent requests worksheet/pack
 2. Server builds Claude prompt with:
 - Child's curriculum data (year + subject + topic list)
 - Performance data (last 20 quiz results for topic)
 - Error patterns (from Error Pattern Detection)
 - Mistake Journal entries
 - Placement level (especially for Arabic)
 3. Claude generates structured JSON:


```

{
  "title": "Carla's Fractions Practice",
  "worked_examples": [...],
  "questions": [...],
  "answers": [...],
  "hints": [...],
  "difficulty_notes": "...
      
```
 4. PDF renderer (ReportLab or puppeteer) applies template:
 - Select template based on subject + type
 - Insert Nuri branding + child's name
 - Handle RTL for Arabic content
 - Generate answer key on separate pages
 5. PDF stored in object storage (S3/Supabase Storage)
 6. URL returned to parent for download/print
 7. PDF auto-expires after 7 days
-

6. Build Phase & Priority

What's Already Planned (from existing spec):

- Parent Highlights (daily note + weekly report) — Phase 5

-
- Snap & Learn (homework photo scanning) — Phase 1

New Parent Hub Features — Suggested Phasing:

Feature	Add to Phase	Effort	Dependencies
Smart Dashboard	Phase 2 (with learning engine)	1 week	Quiz data, mistake journal
Progress Reports (daily/weekly)	Phase 2	1 week	Dashboard data
AI Worksheet Generator	Phase 3 (with gamification)	2 weeks	Claude API, PDF generation
Bilingual Arabic Resources	Phase 3	1 week	Arabic placement, RTL rendering
Smart Weekly Practice Plan	Phase 3	1 week	Spaced repetition engine
Activity Packs	Phase 4	1 week	Worksheet generator
Homework Parent Explainer	Phase 4	1 week	Snap & Learn (Phase 1)
Parent Learning Corner	Phase 5	3-4 days	Curriculum data
Term Reports	Phase 5	1 week	All data sources

Total incremental effort: ~10-11 weeks spread across Phases 2-5, not a separate build.

7. Monetization Impact

The Parent Hub strengthens the subscription model significantly:

Tier	What Parents Get	Impact
Free	Basic dashboard (today's session summary only)	Gives parents a taste — drives upgrade
Standard	Full dashboard, daily notes, 2 worksheets/week, weekly report	Core value
Family/Premium	Unlimited worksheets, activity packs, homework explainer, term reports, practice plans	High-value upsell

Key insight: Parents are the ones who pay. The child uses the tutoring features; the parent uses the Parent Hub. By making the Parent Hub genuinely useful, you justify the subscription to the person holding the credit card.

Competitive pricing advantage: Twinkl charges £5-15/month JUST for static worksheets. Nuri includes personalized worksheets, AI tutoring, gamification, and voice interaction — all in one subscription. Parents get far more value per dollar.

8. Complete Spec Document Map (Updated)

File	Contents
studybuddy-spec.md	Core app, Years 1-6 curriculum, architecture, Nuri mascot, gamification, voice system
studybuddy-features-part2.md	Features 11-20 + DB schema + build phases
studybuddy-features-part3.md	Features 21-35: Error patterns, thinking aloud, emotional adaptation, accessibility
studybuddy-placement-spec.md	Placement Assessment: adaptive English + Arabic assessment at registration
studybuddy-parent-hub-spec.md	Parent Hub: AI worksheet generator, bilingual resources, smart plans, homework explainer, progress reports

Spec created: March 31, 2026

This replaces the simpler "Parent Highlights" feature (Feature 15) with a comprehensive Parent Hub.

The Parent Hub is Nuri's answer to Twinkl — but personalized, bilingual, and AI-powered.